

Ángel Luis Robles Fernández

Postdoctoral Researcher, University of Kansas

Kansas Biological Survey & Center for Ecological Research

a.l.robles.fernandez@gmail.com | [alrobles.github.io](https://github.com/alrobles) | [ORCID 0000-0002-4674-4270](https://orcid.org/0000-0002-4674-4270)

Research profile

Quantitative evolutionary biologist developing demographic species distribution models, machine-learning methods, and open scientific software. My research connects organismal performance, climate variability, species interactions, and population persistence through reproducible computation.

Education

Ph.D., Evolutionary Biology	2025
Arizona State University, School of Life Sciences. Dissertation: <i>Predicting host–parasite interactions from different biodiversity dimensions through machine learning</i> . M.Sc., Physics	
Universidad Veracruzana, Faculty of Physics, Mexico. B.Sc., Physics	2021
Universidad Veracruzana, Faculty of Physics, Mexico.	2017

Academic and professional experience

Postdoctoral Researcher, University of Kansas	Sep 2024–present
Supervisor: Daniel C. Reuman. Co-developer of XSDM, a demographic SDM coupling stochastic population dynamics with climate variability; creator and maintainer of the <code>xsdm</code> R package; developer of Bayesian XSDM methods. Research Assistant, Arizona State University	
Machine-learning frameworks for host–pathogen interactions; co-first-author SPA method linking environmental stability with population genetic diversity; developer of LACS positive–unlabeled literature classification. Senior Automation Analyst, Tenaris Tamsa	Aug 2021–Jul 2025
Machine learning, statistical modeling, databases, dashboards, and industrial process optimization.	Feb 2018–Feb 2020

Selected publications

- * Hernández, N.A.H., Robles-Fernández, Á.L., & Upham, N. (2025). Environmental suitability throughout the late Quaternary explains population genetic diversity. *Ecography*, e07202. Equal contribution.
- * Robles-Fernández, Á.L., Santiago-Alarcon, D., & Lira-Noriega, A. (2022). Wildlife susceptibility to infectious diseases at global scales. *PNAS*, 119, e2122851119.
- * Robles-Fernández, Á.L., Santiago-Alarcon, D., & Lira-Noriega, A. (2021). American mammals susceptibility to dengue according to geographical, environmental, and phylogenetic distances. *Frontiers in Veterinary Science*, 8, 604560.
- * Robles-Fernández, Á.L. & Lira-Noriega, A. (2017). Combining phylogenetic and occurrence information for risk assessment of pest and pathogen interactions with host plants. *Frontiers in Applied Mathematics and Statistics*, 3, 17.
- Berti, E., Robles-Fernández, Á.L., Rosenbaum, B., Peterson, T.A., Soberón, J., & Reuman, D.C. (2026). The impacts of climate variability on the niche concept and distributions of species. *bioRxiv*. doi.
- Krasnov, B.R. et al. (2025). Geographic variation in the determinants of ectoparasite faunas' species richness. *Parasitology*, 152, 745–756.

Research software and data infrastructure

- **XSDM / xsdm**: stochastic-demography SDM with climate variability; R/C++ and RcppParallel.
- **EcoSeek, Genominer, xbioclim, litreview, Sciev**: open web services and workflows for ecological data, population-genetic resources, climate time series, literature synthesis, and structured scientific decisions.
- **Other packages**: `maxentcpp`, `phylopred`, `xniche`, `rosario`, `cofid`, `geotax`, and related R/C++ tools.
- **In development**: `arfun`, with Jorge Soberón, for ancestral niche reconstruction; `ancemb` for Bayesian ancestral protein-embedding reconstruction.

Teaching, mentoring, and awards

- Instructor: Introduction to R; teaching assistant: Quantitative Ecology and Statistical Models for Biology, Arizona State University.
- Invited instructor: graduate ecological niche modeling, INECOL. RStudio Certified Trainer, Tidyverse.
- **GBIF Ebbe Nielsen Challenge, 2022**: second prize for LACS. GBIF Young Researcher Award (2020); American Society of Mammalogists Grant-in-Aid (2023).
- Prepared to teach introductory biology, ecology, evolution, global change biology, quantitative ecology, biogeography,

and R for biologists.

Methods and technical skills

Programming: R, C++17, Python, Shell, SQL, Stan **Computing:** Linux, SLURM, DuckDB, Apache Spark, App-tainer

Methods: machine learning, positive–unlabeled learning, Bayesian inference, SDMs, phylogenetic comparative methods, network analysis

References

Daniel C. Reuman
University of Kansas
reuman@ku.edu

Jorge Soberón
University of Kansas
jsoberon@ku.edu

Diego Santiago-Alarcon
University of South Florida
santiagoalarcon@usf.edu